range of representable values of a number type.

Type	Size in bits	Format	Value range	
			Approximate	Exact
character	8	signed		−128 to 127
		unsigned		0 to 255
	16	UTF-16		0 to 65535
	32	UTF-32		0 to 1114111 (0x10ffff)
integer	16	signed	$\pm 3.27 \cdot 10^{4}$	−32768 to 32767
		unsigned	0 to $6.55 \cdot 10^{4}$	0 to 65535
	32	signed	$\pm 2.14 \cdot 10^{9}$	−2,147,483,648 to 2,147,483,647
		unsigned	0 to $4.29 \cdot 10^{9}$	0 to 4,294,967,295
	64	signed	$\pm 9.22 \cdot 10^{18}$	−9,223,372,036,854,775,808 to 9,223,372,036,854,775,807
		unsigned	0 to $1.84 \cdot 10^{19}$	0 to 18,446,744,073,709,551,615
binary floating-point	32	IEEE-754	• min subnormal: $\pm 1.401{,}298{,}4 \cdot 10^{-45}$ • min normal: $\pm 1.175{,}494{,}3 \cdot 10^{-38}$ • max: $\pm 3.402{,}823{,}4 \cdot 10^{38}$	• min subnormal: ±0x1p−149 • min normal: ±0x1p−126 • max: ±0x1.fffffep+127
	64	IEEE-754	• min subnormal: $\pm 4.940{,}656{,}458{,}412 \cdot 10^{-324}$ • min normal: $\pm 2.225{,}073{,}858{,}507{,}201{,}4 \cdot 10^{-308}$ • max: $\pm 1.797{,}693{,}134{,}862{,}315{,}7 \cdot 10^{308}$	• min subnormal: ±0x1p−1074 • min normal: ±0x1p−1022 • max: ±0x1.fffffffffffffp+1023
	80[note 1]	x86	• min subnormal: $\pm 3.645{,}199{,}531{,}882{,}474{,}602{,}528 \cdot 10^{-4951}$ • min normal: $\pm 3.362{,}103{,}143{,}112{,}093{,}506{,}263 \cdot 10^{-4932}$ • max: $\pm 1.189{,}731{,}495{,}357{,}231{,}765{,}021 \cdot 10^{4932}$	• min subnormal: ±0x1p−16445 • min normal: ±0x1p−16382 • max: ±0x1.fffffffffffffffep+16383
	128	IEEE-754	• min subnormal: $\pm 6.475{,}175{,}119{,}438{,}025{,}110{,}924{,}438{,}958{,}227{,}646{,}552{,}5 \cdot 10^{-4966}$ • min normal: $\pm 3.362{,}103{,}143{,}112{,}093{,}506{,}262{,}677{,}817{,}321{,}752{,}602{,}6 \cdot 10^{-4932}$ • max: $\pm 1.189{,}731{,}495{,}357{,}231{,}765{,}085{,}759{,}326{,}628{,}007{,}016{,}2 \cdot 10^{4932}$	• min subnormal: ±0x1p−16494 • min normal: ±0x1p−16382 • max: ±0x1.ffffffffffffffffffffffffffff p+16383